



Amazon EventBridge - Demo Lab



Objective

Set up an EventBridge rule to trigger an AWS Lambda function when a file is uploaded to an S3 bucket.



Architecture Overview

[S3 Bucket] --> [EventBridge Rule] --> [Lambda Function]



Pre-requisites

- AWS Account
 - IAM Role with required permissions
 - AWS CLI or Management Console
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Step-by-Step Instructions



Step 1: Create an S3 Bucket

1. Go to **S3** in the AWS Console.
2. Click **Create bucket**
 - Name: `eventbridge-demo-bucket`
 - Region: Same as where you will deploy Lambda

3. Leave other options as default and click **Create bucket**
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✓ Step 2: Create a Lambda Function

1. Navigate to **Lambda** → **Create function**
 - Name: `eventbridge-demo-lambda`
 - Runtime: `Python 3.12` (or your preferred language)
 - Role: Choose or create a role with basic Lambda execution and CloudWatch Logs access
2. Replace the default code with:

```
import json

def lambda_handler(event, context):
    print("Received event:")
    print(json.dumps(event, indent=2))
    return {
        'statusCode': 200,
        'body': json.dumps('Event received!')
    }
```

3. Click **Deploy**
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✓ Step 3: Create EventBridge Rule

1. Go to **Amazon EventBridge** → **Rules** → **Create rule**
2. **Name:** `s3-upload-trigger-rule`
3. **Event bus:** `default`

4. **Rule type:** Event pattern

5. **Define pattern:**

- Select **AWS services**
- Service name: `S3`
- Event type: `Object Created`
- Bucket name: `eventbridge-demo-bucket`

6. **Target:**

- Target type: `AWS service`
- Select: `Lambda function`
- Function: `eventbridge-demo-lambda`

7. Click **Create**

✓ **Step 4: Upload a File to S3**

1. Go to your S3 bucket
 2. Click **Upload**
 3. Upload any test file (e.g., `test.txt`)
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✓ **Step 5: Verify Execution**

1. Navigate to **CloudWatch Logs**
2. Find the log group for your Lambda: `/aws/lambda/eventbridge-demo-lambda`

3. Check the latest log stream
4. You should see the S3 event details in the Lambda logs.



Expected Output

The Lambda logs will show event metadata like:

```
{
  "version": "0",
  "id": "abc123",
  "detail-type": "Object Created",
  "source": "aws.s3",
  "account": "111122223333",
  "time": "2025-08-01T10:00:00Z",
  "region": "us-east-1",
  "resources": [
    "arn:aws:s3:::eventbridge-demo-bucket"
  ],
  "detail": {
    "bucket": {
      "name": "eventbridge-demo-bucket"
    },
    "object": {
      "key": "test.txt"
    }
  }
}
```



Pricing Notes

- **EventBridge:** \$1.00 per million events published.
 - **S3 + Lambda:** Minimal costs if using free-tier eligible account and small test files.
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